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YouTube Recommendation Algorithm: A/B Experiment Analysis

January 2024 · 30-day experiment · 5,000 users · Dataset: synthetically generated

The problem

YouTube's recommendation system has a built-in tension that I found genuinely interesting to work through. Short-form entertainment content dominates most feeds. It drives strong click-through rates and keeps sessions active. But there's a hypothesis that long-form educational content creates deeper, more durable engagement: users who watch tutorials and explainers tend to stay longer and come back more often.

The product team proposed a new algorithm that surfaces more educational content. Before shipping it, they needed answers to two questions: does it actually increase engagement, and does it do that without hurting the short-form and entertainment creators who depend on the platform for their livelihood? Those two things are in tension, and that tension is exactly what makes this an interesting experiment to analyze.

What I did

I ran a full end-to-end product analytics case study on this. I'll walk through the main pieces:

Experiment design and KPI framework. I defined the success criteria before touching any data: one primary metric (avg daily watch time per user), secondary metrics to understand the mechanism (session duration, retention, videos per session), and two guardrail metrics for creator ecosystem health. Defining guardrails upfront is the thing that prevents a team from shipping something that looks great on the headline number but quietly breaks something important.

Statistical testing. Welch's two-sample t-test on the primary metric, bootstrap confidence intervals, Cohen's d for effect size. I also applied CUPED, a variance reduction technique that uses each user's pre-experiment watch time as a covariate. It's standard at Google and Netflix for making experiments more sensitive without increasing sample size. The CUPED result confirmed the raw finding, which gave me more confidence in it.

Segmentation and novelty check. I broke down the treatment effect by device, age group, and account type. I also measured the lift week by week to check for novelty effects, where users engage more with anything new in week 1 and then revert. The week-by-week pattern actually showed the opposite: the lift grew over time, which is a stronger signal than a flat effect.

Root cause decomposition. I decomposed the watch time gain into frequency (users opening the app more often) versus duration (users staying longer per session). This matters for the engineering team: it tells them which lever is actually moving.

Key numbers

Metric	Control	Treatment	Change
Avg daily watch time	12.0 min	13.5 min	+1.5 min (+13%)
Session duration	33.7 min	35.3 min	+1.6 min
Videos per session	7.3	5.7	-1.6
Week-4 retention	73.9%	77.8%	+3.9 pp

p-value: 0.0000 · Cohen's d: 0.21 · Bootstrap 95% CI: [1.14, 1.95] min/day · CUPED variance reduction: 13%

The thing that concerned me

The primary metric result is clean: watch time is up, the effect is significant, and it holds across all four weeks. That part was straightforward.

What I spent more time on was the creator ecosystem numbers. The new algorithm shifts the content mix significantly:

Content type	Control	Treatment	Change
Short entertainment	50.2%	29.8%	–20.4 pp
Long entertainment	29.8%	28.2%	–1.6 pp
Long educational	12.0%	32.1%	+20.1 pp
Short educational	7.9%	10.0%	+2.1 pp

> Short-form entertainment creators are receiving ~20 percentage points fewer impressions. That's not a rounding error; it's a significant hit to a large segment of the creator base.

I set a guardrail threshold of -5 pp before running any analysis. The short-form impression drop is four times that. The algorithm is doing exactly what it was designed to do, surfacing more educational content, but it's gone too far in the current form. Shipping this as-is would hurt a meaningful segment of creators who depend on YouTube for their livelihood.

My recommendation

> **Don't ship in the current form.** Recalibrate the content blend, re-run the experiment, then ship.

The core idea is sound. The engagement lift is real and durable, and the week-4 retention improvement is arguably the most compelling finding: it suggests users who stick with the new algorithm are genuinely more engaged, not just novelty-curious.

But the creator impact is too large to ignore. My specific suggestion: work with the algorithm team to add a diversity constraint that limits how aggressively educational content is promoted. Target a max 8-10 pp shift in short-form impression share, then re-test. If the primary metric lift survives at a reduced magnitude and the guardrail passes, it's a clear ship.

After launch, I'd want creator-level impression tracking as a standing dashboard, not just a pre-launch check. Recommendation system effects can compound over time in ways a 30-day experiment won't fully capture.

Full analysis (Jupyter notebook with all code and charts) available on request. Dataset is synthetically generated; all analytical work is original.